

RAM KRISHNA TRIPATHI

☎ +91-8780518300 ✉ tripathiramkrishna16@gmail.com [in LinkedIn](#) [Github](#) [LeetCode](#) [Codeforces](#)

Education

Indian Institute of Information Technology, Gwalior

Aug 2023 – Jun 2028

Integrated Bachelors and Masters in Information Technology

CGPA: 8.77

Aatmdeep Vidyalaya, Gorakhpur

June 2022

All India Senior School Certificate Examination

Percentage: 94.4%

Experience

AI Trainer | **Outlier AI** | Remote

Oct 2024 – Oct 2025

- Evaluated Large Language Models (LLMs) across **20+ AI evaluation projects**, designing prompts to identify reasoning and instruction-following failures under project-specific guidelines.
- Completed **370+ evaluation tasks**, documenting model failure cases and submitting detailed analyses with corrected reference solutions for quality assessment.

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, FastAPI, Tailwind CSS, Shadcn UI

Databases: PostgreSQL, MongoDB, MySQL, pgvector

AI/ML: Retrieval-Augmented Generation (RAG), LangChain, Vector Search, Semantic Search, Cohere API, Gemini API

Tools & Platforms: Git, CMake, Prisma ORM, Supabase, NextAuth.js, Chrome Extension (Manifest V3), LaTeX

Projects

Kafka Lite – Kafka-inspired Persistent Message Broker 🔄

June 2026

- Built a **Kafka-inspired persistent message broker** in **C++17** featuring TCP-based client-server communication, append-only log storage, topic partitioning, and crash recovery through durable metadata and log reconstruction.
- Implemented **Consumer Groups** with automatic partition assignment, heartbeat-based failure detection, dynamic rebalancing, persistent consumer offsets, producer acknowledgements, configurable retention, and batch publishing.
- Benchmarked the broker, achieving **331K+ messages/sec** peak throughput, **0.46 ms** publish latency, and recovery from **1M messages in 0.5 sec**, with metrics APIs and a monitoring dashboard.
- Tech Stack:** C++17, STL, TCP Sockets, Multi-threading, CMake, nlohmann::json, Python, Mermaid, Git.

DocSense AI – AI-powered Knowledge Search Platform 🔄🌐

April 2026

- Built an **AI-powered knowledge platform** enabling semantic search and Retrieval-Augmented Generation (RAG) over **PDFs, DOCX, TXT, websites, GitHub repos, and YouTube transcripts** with source-backed responses.
- Developed **DocSense Companion**, a **Chrome Extension (Manifest V3)** for one-click webpage ingestion, instant content summarization, and seamless synchronization with the DocSense AI knowledge base.
- Implemented **vector embeddings, hybrid search, document chunking, collection management, and inline citations** to deliver accurate, context-aware conversational search across personal knowledge bases.
- Tech Stack:** Next.js 15, React 19, TypeScript, FastAPI, PostgreSQL, pgvector, Prisma ORM, Cohere API, Gemini API, Tailwind CSS, Chrome Extension (Manifest V3), Git.

Brick Basket – Home Construction Management Platform 🔄🌐

May 2026

- Built a **B2C construction management platform** using Next.js 15 for project tracking, milestone management, client communication, and **daily photo updates** to enhance client transparency.
- Designed a secure multi-role system with **Role-Based Access Control (RBAC)**, NextAuth.js v5 authentication, PostgreSQL, Prisma ORM, Supabase Storage, and an automated **waterfall payment distribution** engine.
- Tech Stack:** Next.js 15, React 19, TypeScript, PostgreSQL, Prisma ORM, NextAuth.js v5, Supabase, Tailwind CSS, Shadcn UI, Framer Motion.

Relevant Courseworks

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems

Systems: Operating Systems, Computer Networks, Computer Organization & Architecture

Achievements

- Selected as a **National Semi-Finalist** in **Flipkart Grid 7.0** 🏆, one of India's premier Engineering competitions.
- Solved **500+ DSA problems** across coding platforms including **LeetCode**, **CodeChef**, and **Codeforces**.
- Achieved a **1750+ rating** on **LeetCode** and secured **Global Rank 848** among **30,290 participants** in **LeetCode Biweekly Contest 481**.